Discussion / Exercise session

Shiquin's questions.

1. Is $\langle x^2+y^2+1\rangle \subseteq \mathbb{R}[x,y]$ maximal?
2. Are there ideals $I \subseteq k[x_1,\dots,x_n]$ that require more than n polynomials to be generated?
3. Discussion on different types of ideals (maximal, prime, but there are also more!)

$n=1$ is clear.
We're talking about $n \geq 2$.

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(1) $I := \langle x^2+y^2+1\rangle$. Let me introduce another one: $J := \langle x^2+y^2+1, y\rangle$

$I \subsetneq J$: $I \subseteq J$ is clear (we've got another added to J) and $y \notin I$ (because every polynom. in I there is nonzero has degree at least two).

Is $J = \mathbb{R}[x,y]$ or <u>not</u>? In fact, $J \neq \mathbb{R}[x,y]$, so I is not a maximal ideal!

It seems that $x \notin \langle x^2+y^2+1, y\rangle$.

With the Gröbner bases, we can decide this by computing the Gröbner basis and dividing x by the basis.

We can try doing this manually.

We do it "manually" and just argue by contradiction. Assume, we had $x \in \langle x^2+y^2+1, y\rangle$.

Then $x = (x^2+y^2+1)\,p(x,y) + y\cdot q(x,y)$ with some $p,q \in \mathbb{R}[x,y]$.

Quotient ring approach

- $\mathbb{R}[x]/\langle x^2+1\rangle$
- $\mathbb{R}[x]/\langle x\rangle$
- $\mathbb{R}[x,y]/\langle x^2+y^2+1, y\rangle \cong \mathbb{R}[x]/\langle x^2+1\rangle$

Let's try fixing $y=0$:

$$x = (x^2+1)\cdot p(x,0) \leftarrow \text{identity in } \mathbb{R}[x]$$

Two possibilities: either $p(x,0) \equiv 0 \in \mathbb{R}[x]$; then we have $x=0$ in $\mathbb{R}[x]$, which is false.

or $p(x,0) \neq 0$; then $\deg(x) = 1 < \deg((x^2+1)\cdot p(x,0))$, which is also false.

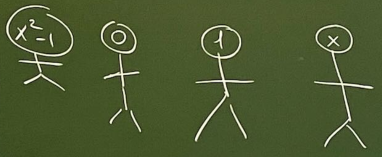
Solve equation $k=\mathbb{R}$

$x^2-5x+6=0$

$V_{\mathbb{R}}(x^2-5x+6)$

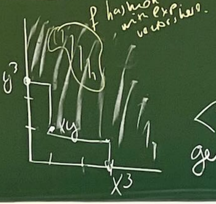
x is not always the formal variable in $k[x]$ or $k[x,y]$ or etc.; sometimes I used it for something else (like in CRT). The reason: x is also frequently used to be the input of a map, and in the CRT we did have a map. Please, don't be confused!

Question. What does $k[x]$ mean? In particular $\mathbb{R}[x]$.

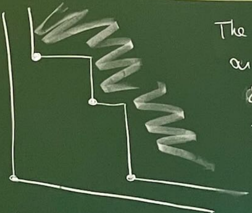


$$x^2-1 = (x-1)(x+1)$$

$$0\cdot x^3 + 1\cdot x^2 + (-1)\cdot x^0$$



(2)



The answer is $\boxed{\text{yes}}$: upper bound on the number of generators sufficient to generate I, an ideal in $k[x_1,\dots,x_n]$ when $n \geq 2$. Look at the monomial ideals.

$\langle x^3, xy, y^3\rangle \leftarrow$ to show that this one cannot be generated by two polynomials needs some work.

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$$\underbrace{\langle x^3, xy, y^3\rangle}_{\substack{\text{not maximal,}\\ \text{requires } 3\,(>n=2)\\ \text{generators}}} \subsetneq \underbrace{\langle x,y\rangle}_{\substack{\text{maximal,}\\ \text{has only } 2=n\\ \text{generators}}} \subsetneq k[x,y]$$

maximal $\Rightarrow$ $\underbrace{\text{prime}}_{\substack{\text{at most } n\\ \text{generators are sufficient.}}}$

An ideal I is the whole ring R iff $1 \in I$.
So, to test if $\langle x,y\rangle \neq k[x,y]$ it's always enough to test $1 \notin \langle x,y\rangle$.

i.e. $1 = x\,p(x,y) + y\,q(x,y)$

$\Rightarrow 1 = 0\cdot p(0,0) + 0\cdot q(0,0) \Rightarrow$ Contradiction